

Conical Intersections as Typed Spectral Singularities

Eigenprojectors, holonomy, nonadiabatic geometry, and observation maps

Stas

Independent Research Program; ORCID: 0009-0000-2294-705X

Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0. It is an application-layer reference migrated to GO Core v0.2. It does not propose a new electronic-structure method, nonadiabatic dynamics algorithm, conical-intersection theorem, or optical-caustic equivalence. The internal contribution is a typed separation of nuclear coordinates, energy-valued branching coordinates, gauge-invariant projectors, local eigenvector sections, dynamics, and finite-resolution observation.

Abstract

Let q denote physical nuclear coordinates and let $H_e(q)$ be an electronic Hamiltonian family with an isolated two-state spectral cluster. Near a generic real two-state degeneracy, the linear branching-plane model is

$$H_{\text{eff}}(q) = E_0(q)I + x(q)\sigma_z + y(q)\sigma_x, \quad x = g_\mu \delta q^\mu, \quad y = h_\mu \delta q^\mu,$$

where x, y are energies, not nuclear lengths. The seam is the regular preimage $x = y = 0$, hence has codimension two when $\text{rank } D_q(x, y) = 2$. On the punctured branching plane,

$$E_\pm = E_0 \pm r, \quad P_\pm = \frac{1}{2} \left(I \pm \frac{x\sigma_z + y\sigma_x}{r} \right), \quad r = \sqrt{x^2 + y^2}.$$

The unordered spectrum and the two-state cluster remain meaningful at the degeneracy, but neither rank-one projector has a unique limit there. Eigenvectors are only gauge-dependent local sections. A loop of winding number w has real-line holonomy $(-1)^w$ and Berry phase πw modulo 2π . In the energy coordinates, the gauge-invariant quantum metric is

$$g^{(E)} = \frac{1}{4r^4} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}, \quad \text{Tr } g^{(E)} = \frac{1}{4r^2}.$$

Its pullback by $D_q(x, y)$ supplies the correctly dimensioned physical-coordinate coupling. A closing gap invalidates a uniform adiabatic estimate, but does not determine a transition probability; that requires a nuclear path, speed, initial state, and dynamical model. Finally, finite spectral resolution can certify only an unresolved near-degeneracy unless uncertainty and identifiability conditions are supplied. The strict object is a spectral-discriminant singularity. “Observation caustic” is retained only as a qualified program-level analogy.

Contents

1	Scope and typed setting	2
1.1	Electronic family and spectral cluster	2
1.2	Physical and energy-valued coordinates	2

2 Spectral discriminant and codimension	3
3 Projectors, eigenlines, and gauge	3
4 Berry holonomy without a global eigenvector	4
5 Derivative coupling and quantum metric	4
6 Born-Oppenheimer reduction and dynamics firewall	5
7 Observation chain and finite-resolution identifiability	6
8 Gap-opening control and reproducible benchmarks	6
9 Protocol requirements and claim register	7
References	8

1 Scope and typed setting

1.1 Electronic family and spectral cluster

Let $U \subset \mathbb{R}^f$ be a nuclear-coordinate chart. For the strict finite-dimensional reference, $H_e : U \rightarrow \text{Herm}(n)$ is C^2 . The two electronic levels of interest form a cluster separated from the remaining spectrum by a positive external gap on a neighborhood $V \subset U$. The Riesz projector onto that cluster is

$$P_{\text{cl}}(q) = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_e(q))^{-1} dz, \quad (1)$$

where one fixed contour Γ encloses precisely the two cluster eigenvalues. Under these hypotheses P_{cl} is C^2 , even when the two eigenvalues cross internally.

For an unbounded electronic operator, a common dense domain, self-adjointness, resolvent regularity, and an isolated spectral cluster must be stated separately. The finite matrix reference does not silently prove those operator-theoretic hypotheses.

1.2 Physical and energy-valued coordinates

Choose $q_* \in U$, set $\delta q = q - q_*$, and write the linear two-state reduction after a fixed local basis choice as

$$H_{\text{eff}}(q) = E_0(q)I + x(q)\sigma_z + y(q)\sigma_x + O(\|\delta q\|^2), \quad x = g_\mu \delta q^\mu, \quad y = h_\mu \delta q^\mu. \quad (2)$$

Here q^μ is a physical nuclear coordinate, while g_μ, h_μ are energy-gradient covectors. Thus

$$[q^\mu] = L, \quad [g_\mu] = [h_\mu] = EL^{-1}, \quad [x] = [y] = E$$

in the declared length chart. A mass-weighted or curvilinear chart is allowed only with its own coordinate dimensions and Jacobian.

Boundary 1.1 (Coordinate firewall). The notation $H = Q_1\sigma_z + Q_2\sigma_x$ is dimensionally valid only if Q_1, Q_2 are declared energy-valued branching coordinates. It is not valid when the same symbols denote unscaled nuclear lengths.

2 Spectral discriminant and codimension

The exact benchmark drops $E_0 I$ and higher-order terms:

$$H(x, y) = x\sigma_z + y\sigma_x = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \quad r = \sqrt{x^2 + y^2}. \quad (3)$$

Proposition 2.1 (Conical spectrum). *For $(x, y) \neq (0, 0)$,*

$$E_{\pm} = \pm r, \quad \Delta := E_+ - E_- = 2r. \quad (4)$$

The repeated eigenvalue occurs precisely at $x = y = 0$.

Proof. $\det(H - EI) = E^2 - x^2 - y^2$, so $E = \pm r$. Equality of the two eigenvalues is equivalent to $r = 0$. $\square$

Let $F = (x, y) : U \rightarrow \mathbb{R}^2$ and $S = F^{-1}(0)$.

Proposition 2.2 (Generic real seam). *If $\text{rank } D_q F(q_*) = 2$, then S is a C^1 submanifold of codimension two near q_* . Its tangent space is*

$$T_{q_*} S = \ker D_q F(q_*).$$

Proof. This is the regular-value theorem applied to F . $\square$

Boundary 2.3 (Symmetry class matters). A general complex Hermitian two-state matrix contains three independent traceless coefficients; its generic degeneracy has codimension three. Symmetry-enforced crossings, spin-orbit coupling, Kramers degeneracy, and higher multiplicity require different discriminants. Codimension two is asserted only for the declared real two-state class with a transverse parameter map.

3 Projectors, eigenlines, and gauge

On $\mathbb{R}^2 \setminus \{0\}$, define the gauge-invariant rank-one projectors

$$P_{\pm}(x, y) = \frac{1}{2} \left(I \pm \frac{x\sigma_z + y\sigma_x}{r} \right). \quad (5)$$

Proposition 3.1 (No rank-one extension through the seam). *Neither P_+ nor P_- has a continuous extension to $(x, y) = (0, 0)$. The cluster projector $P_{\text{cl}} = P_+ + P_- = I_2$ does extend smoothly.*

Proof. Along $y = 0$, the limits of P_- as $x \rightarrow 0^+$ and $x \rightarrow 0^-$ are respectively $\text{diag}(0, 1)$ and $\text{diag}(1, 0)$. They differ. The sum is identically I_2 . $\square$

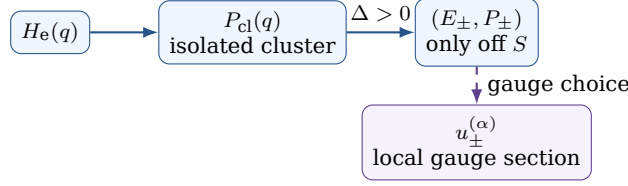
Write $x = r \cos \phi$, $y = r \sin \phi$. One real local lift is

$$u_-(\phi) = \begin{pmatrix} -\sin(\phi/2) \\ \cos(\phi/2) \end{pmatrix}, \quad u_+(\phi) = \begin{pmatrix} \cos(\phi/2) \\ \sin(\phi/2) \end{pmatrix}. \quad (6)$$

It satisfies $u_{\pm}(\phi + 2\pi) = -u_{\pm}(\phi)$. A local eigenvector is not the output of a global gauge-invariant observation map. Under

$$u_a(q) \mapsto e^{i\chi_a(q)} u_a(q),$$

the projector $P_a = |u_a\rangle\langle u_a|$ is unchanged.



4 Berry holonomy without a global eigenvector

Let $C : [0, 1] \rightarrow \mathbb{R}^2 \setminus \{0\}$ be a closed loop and let $w(C, 0) \in \mathbb{Z}$ be its winding number around the origin.

Proposition 4.1 (Real-line holonomy). *For the model equation (3), parallel transport of either real eigenline around C has holonomy*

$$\text{Hol}(C) = (-1)^{w(C,0)}, \quad \gamma(C) = \pi w(C, 0) \pmod{2\pi}. \quad (7)$$

Proof. The polar angle changes by $2\pi w$. Equation (6) therefore changes the real lift by $(-1)^w$. The ray returns to itself, while the lifted line has the stated holonomy. $\square$

A gauge-invariant discrete diagnostic for a sufficiently fine loop $q_0, \dots, q_{N-1}$, with nonzero neighboring overlaps, is

$$W_N = \prod_{j=0}^{N-1} \frac{\langle u_j, u_{j+1} \rangle}{|\langle u_j, u_{j+1} \rangle|}, \quad u_N = u_0, \quad \gamma_N = \arg W_N. \quad (8)$$

Local phase factors telescope in the closed product. In a real gauge, $W_N = -1$ for odd winding and $+1$ for even winding.

Boundary 4.2 (Bundle and curvature boundary). Over a punctured plane, a complex eigenline can be represented by a single-valued complex section, but the Berry connection may have nontrivial holonomy. In the real model, the real line over a loop of odd winding is the nontrivial real line bundle. The smooth Berry curvature vanishes away from the origin in this two-dimensional real model; the phase must not be attributed to an ordinary nonzero local curvature density on the punctured plane.

5 Derivative coupling and quantum metric

For local orthonormal sections, define

$$d_{ab}^{(q)} = \langle u_a, \nabla_q u_b \rangle.$$

For $a \neq b$, a gauge change multiplies d_{ab} by a phase; therefore d_{ab} itself is gauge covariant, while its norm and the quantum metric are gauge invariant.

In the energy coordinates $\xi = (x, y)$, the gauge equation (6) gives

$$d_{+-}^{(E)} = -\frac{1}{2} \nabla_\xi \phi = \frac{1}{2r^2} \begin{pmatrix} y \\ -x \end{pmatrix}, \quad \|d_{+-}^{(E)}\| = \frac{1}{2r}. \quad (9)$$

Its dimension is E^{-1} , not L^{-1} .

The rank-one quantum metric is

$$g_{\mu\nu}^{(E)} = \text{Re} \langle \partial_\mu u_-, (I - P_-) \partial_\nu u_- \rangle = \frac{1}{2} \text{Tr}(\partial_\mu P_- \partial_\nu P_-) = \frac{1}{4r^4} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}_{\mu\nu}. \quad (10)$$

Hence

$$\text{Tr } g^{(E)} = \frac{1}{4r^2} = \|d_{+-}^{(E)}\|^2. \quad (11)$$

Let $J = D_q(x, y)$, with entries of dimension EL^{-1} in the declared length chart. The physical-coordinate quantities are the pullbacks

$$d_{+-}^{(q)} = J^\top d_{+-}^{(E)}, \quad g^{(q)} = J^\top g^{(E)} J, \quad (12)$$

so $[d_{+-}^{(q)}] = L^{-1}$ and $[g^{(q)}] = L^{-2}$. The scalar $1/(2r)$ is therefore not a coordinate-independent physical coupling magnitude; it belongs to the Euclidean metric of the chosen energy-coordinate plane.

6 Born-Oppenheimer reduction and dynamics firewall

For a local adiabatic expansion

$$\Psi(q, r_e) = \sum_a \chi_a(q) u_a(q; r_e),$$

the nuclear kinetic operator contains

$$(T_N)_{ab} = - \sum_A \frac{\hbar^2}{2M_A} \left[\delta_{ab} \nabla_A^2 + 2d_{ab}^{(A)} \cdot \nabla_A + \tau_{ab}^{(A)} \right], \quad \tau_{ab}^{(A)} = \langle u_a, \nabla_A^2 u_b \rangle. \quad (13)$$

Under a coordinate-dependent unitary change within the retained cluster, covariance belongs to the complete matrix operator. Discarding the off-diagonal terms is an approximation whose uniform validity fails near an internal gap closure.

For a prescribed path $q(t)$, a local adiabatic diagnostic is

$$\eta_{ab}(t) = \frac{\hbar \left| \langle u_a, \dot{H}_e u_b \rangle \right|}{|E_b - E_a|^2}. \quad (14)$$

The condition $\eta_{ab} \ll 1$ is a model-dependent sufficiency diagnostic, not a universal error theorem. At a gap closure, the uniform-gap hypothesis of standard adiabatic estimates fails.

Model 6.1 (Landau-Zener control). For

$$H_{\text{LZ}}(t) = vt \sigma_z + b \sigma_x, \quad v > 0, \quad b > 0,$$

with evolution from $t = -\infty$ to $+\infty$, the exact diabatic survival probability is

$$P_D = \exp \left(-\frac{\pi b^2}{\hbar v} \right). \quad (15)$$

This control has minimum gap $2b$, but changing the sweep rate v changes P_D . Thus the static gap, Berry phase, or derivative coupling alone does not determine a molecular branching ratio. Wavepacket shape, masses, path, velocity, decoherence, additional states, and the measurement channel are separate inputs.

7 Observation chain and finite-resolution identifiability

The strict factorization is

$$(H_{\text{mol}}, \Psi) \xrightarrow{\mathcal{S}_{\text{cl}}} (H_{\text{cl}}(q), \chi(q)) \xrightarrow{\mathcal{P}_{\text{ad}}} (E_{\pm}(q), P_{\pm}(q), \chi_{\pm}(q)) \xrightarrow{\mathcal{O}_{\lambda}} Z_{\lambda} \xrightarrow{\mathcal{K}_{\lambda}} \text{Law}(Y_{\lambda} | Z_{\lambda}) \xrightarrow{\hat{\Theta}_{\lambda}} \hat{\Theta}. \quad (16)$$

The rank-one adiabatic splitting $\mathcal{P}_{\text{ad}}$ is defined only off the seam. A local eigenvector section is an additional gauge choice, not an observation. A spectroscopic or product channel $\mathcal{O}_{\lambda}$ does not generally reveal the full projector field or Berry holonomy.

Definition 7.1 (Spectral-discriminant singularity). Let $\mathcal{D} \subset \text{Herm}(n)$ be the repeated-eigenvalue discriminant. A conical intersection of the declared real two-state class is a transverse point or seam in

$$S = H_{\text{eff}}^{-1}(\mathcal{D})$$

for which the ordered rank-one eigenprojector map fails to extend through S and the leading eigenvalue splitting is conical.

This is not the same as requiring $\text{rank } DH_{\text{eff}}$ to drop. The singular map is the eigen-decomposition at the discriminant, not necessarily the parameter-to-Hamiltonian map. For a classical optical caustic, the reusable criterion is rank loss of a ray projection; a determinant criterion is valid only in a square local chart.

Boundary 7.2 (Qualified use of “observation caustic”). The term may be used as a GO analogy for singularity of a reduced description. It is not a theorem that conical intersections and optical caustics are equivalent singularities, and no Arnold classification of Lagrangian-map caustics is imported here.

Suppose measured energies are

$$Y_{\pm} = E_{\pm} + \varepsilon_{\pm}, \quad \mathbb{E}[\varepsilon_{\pm}] = 0.$$

Then the measured gap $\hat{\Delta} = Y_+ - Y_-$ has variance

$$\sigma_{\hat{\Delta}}^2 = \sigma_+^2 + \sigma_-^2 - 2 \text{Cov}(\varepsilon_+, \varepsilon_-). \quad (17)$$

With instrumental resolution $\epsilon_E \geq 0$, a rule such as

$$|\hat{\Delta}| \leq \kappa \sigma_{\Delta} + \epsilon_E \quad (18)$$

classifies an *unresolved near-degeneracy*. It does not prove an exact degeneracy, conical topology, odd Berry holonomy, or a nonadiabatic transition. Those conclusions require additional sampling, model class, path, and identifiability assumptions.

8 Gap-opening control and reproducible benchmarks

The complex Hermitian control

$$H_{\delta}(x, y) = x\sigma_z + y\sigma_x + \delta\sigma_y, \quad \delta \neq 0, \quad (19)$$

has

$$E_{\pm}^{\delta} = \pm \sqrt{x^2 + y^2 + \delta^2}, \quad \Delta_{\delta} = 2\sqrt{r^2 + \delta^2} \geq 2|\delta|. \quad (20)$$

It opens a gap on the x, y plane by leaving the real-symmetric class. It is therefore a symmetry-breaking control, not the same real conical-intersection model with a harmless numerical regularizer.

For the positively oriented circle $x = R \cos \phi$, $y = R \sin \phi$, and the orientation convention used in the release, the lower-state phase modulo 2π has magnitude

$$\gamma_\delta(R) = \pi \left(1 - \frac{\delta}{\sqrt{R^2 + \delta^2}} \right). \quad (21)$$

It tends to π as $\delta \rightarrow 0^+$ and to zero as $\delta/R \rightarrow +\infty$; unlike the real sign holonomy, it is not quantized for $\delta \neq 0$.

Benchmark	Exact value	Purpose
$H(3, 4)$, energy units	$E_\pm = \pm 5$, $\Delta = 10$	Conical spectrum and dimensions
$r = 0.25$, energy units	$\ d_{+-}^{(E)}\ = 2$	Energy-coordinate coupling scaling
Unit circle, winding 1	$W_N = -1$, $\gamma = \pi$	Gauge-invariant discrete holonomy
Unit circle, winding 2	$W_N = +1$, $\gamma = 0 \bmod 2\pi$	Parity of real-line holonomy
$R = 1, \delta = 1$	$\gamma_\delta = \pi(1 - 1/\sqrt{2})$	Nonquantized gapped control
$\hbar v = b^2$	$P_D = e^{-\pi}$	Dimensionless dynamical exponent

Table 1: Reference values implemented by the P8 regression suite.

The executable release additionally checks projector idempotence and orthogonality, absence of a rank-one limit, cluster smoothness, gauge invariance of the overlap product, quantum-metric identities, coordinate pullback dimensions, the Landau-Zener formula, covariance in equation (17), and zero-gap guards.

9 Protocol requirements and claim register

A complete conical-intersection observation record states at least:

1. nuclear-coordinate chart, coordinate dimensions, reference geometry, and frame;
2. electronic Hamiltonian class, domain assumptions, basis, symmetry class, and retained spectral cluster;
3. external cluster gap, active-state multiplicity, and degeneracy/discriminant convention;
4. physical-to-branching Jacobian $D_q(x, y)$ and its units;
5. whether outputs are unordered eigenvalues, ordered energies, projectors, local eigenvectors, or a diabatic matrix;
6. gauge convention, loop orientation, winding rule, and overlap threshold for a discrete Berry phase;
7. nuclear path, speed, masses, initial state, and dynamical approximation for every transition claim;
8. spectroscopic or product observation map, resolution, stochastic kernel, covariance, and detection threshold;
9. estimator, uncertainty statement, sampling region, and identifiability class;
10. explicit boundary between exact normal-form results, asymptotic diagnostics, numerical evidence, and chemical interpretation.

The strict program-level conclusion is that a conical intersection is a transverse encounter with a spectral discriminant for which the individual adiabatic rank-one splitting becomes singular while the retained two-state cluster can remain regular. Gauge, dynamics, and measurement are separate typed layers.

Claim	Status	Scope
Double-cone spectrum	Proposition	Exact real two-state normal form
Codimension-two seam	Proposition	rank $D_q(x, y) = 2$, real two-state class
No rank-one projector extension	Proposition	Punctured branching plane approaching the origin
Berry sign holonomy	Proposition	Closed loop with declared winding, real eigenline
$\text{Tr } g^{(E)} = 1/(4r^2)$	Proposition	Euclidean energy-coordinate metric
Gap closure fixes transition probability	Rejected	Dynamics and initial data are additionally required
Finite unresolved gap proves a CI	Rejected	Resolution alone does not identify topology
CI equals an optical caustic	Rejected	Only a qualified observation-reduction analogy
New quantum-chemical method or theorem	Rejected	Not claimed

References

- [1] M. Born and R. Oppenheimer, “Zur Quantentheorie der Molekeln,” *Annalen der Physik* 389, 457–484 (1927), doi:10.1002/andp.19273892002.
- [2] H. C. Longuet-Higgins, U. Oepik, M. H. L. Pryce, and R. A. Sack, “Studies of the Jahn-Teller effect. II. The dynamical problem,” *Proceedings of the Royal Society A* 244, 1–16 (1958), doi:10.1098/rspa.1958.0022.
- [3] G. Herzberg and H. C. Longuet-Higgins, “Intersection of potential energy surfaces in polyatomic molecules,” *Discussions of the Faraday Society* 35, 77–82 (1963), doi:10.1039/DF9633500077.
- [4] C. A. Mead and D. G. Truhlar, “On the determination of Born-Oppenheimer nuclear motion wave functions including complications due to conical intersections and identical nuclei,” *Journal of Chemical Physics* 70, 2284–2296 (1979), doi:10.1063/1.437734.
- [5] M. V. Berry, “Quantal phase factors accompanying adiabatic changes,” *Proceedings of the Royal Society A* 392, 45–57 (1984), doi:10.1098/rspa.1984.0023.
- [6] C. Zener, “Non-adiabatic crossing of energy levels,” *Proceedings of the Royal Society A* 137, 696–702 (1932), doi:10.1098/rspa.1932.0165.
- [7] T. Kato, *Perturbation Theory for Linear Operators*, Springer, corrected printing of the second edition, 1995.
- [8] D. R. Yarkony, “Diabolical conical intersections,” *Reviews of Modern Physics* 68, 985–1013 (1996), doi:10.1103/RevModPhys.68.985.